

Statistical Process Control & Yield-Excursion Analysis of a Wafer Process

Izaiah Thigpen

Semiconductor Process Data Analysis · Python / pandas

Objective

Statistical process control (SPC) monitors a manufacturing line by tracking process parameters against control limits so that excursions are caught before they cost yield. This analysis applies SPC to a 5,000-wafer dataset spanning six process parameters and, using the small population of defective wafers, tests whether yield loss coincides with process-window excursions — the kind of root-cause link a quality engineer uses to prioritize which parameters to tighten.

Method

Individuals / moving-range (I-MR) control limits were computed for each parameter, estimating the process standard deviation from the mean moving range ($\sigma = \text{mean MR} / 1.128$) and setting 3σ control limits about the mean. Wafers outside the limits were flagged as out-of-control. To characterize the defects, each defective wafer's parameters were expressed as z-scores relative to the population, and the mean z-score per parameter (the "excursion fingerprint") was computed to reveal which parameters run abnormally on failing wafers.

Results

Parameter	Mean	σ (I-MR)	LCL (3σ)	UCL (3σ)	#OOC	Defect z
Temperature (°C)	450.1	15.10	404.8	495.4	14	+1.37
Pressure (Torr)	759.7	30.63	667.8	851.6	12	-2.30
Gas flow (sccm)	120.1	9.90	90.4	149.8	21	-0.30
Etch rate (nm/min)	95.1	8.07	70.9	119.3	11	+1.38
Voltage (V)	5.0	0.39	3.8	6.2	14	+0.43
Current (mA)	20.0	2.01	14.0	26.0	11	+0.25

Wafer Process SPC & Yield-Excursion Analysis (5,000 wafers)

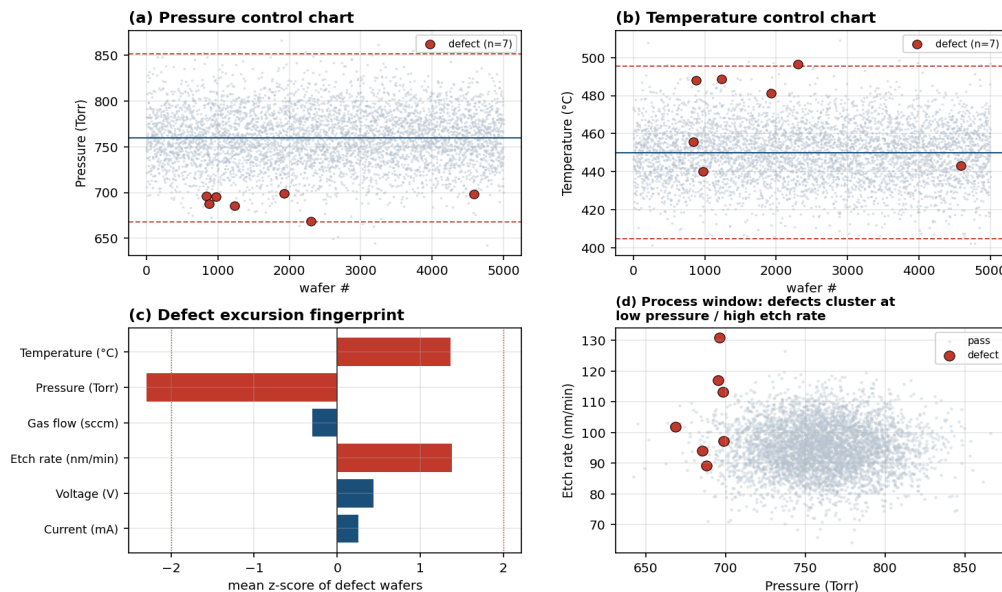


Figure 1. (a,b) Individuals control charts for pressure and temperature with 3σ limits; defective wafers (red) sit at the limits. (c) Excursion fingerprint of the defect wafers. (d) Process-window scatter: defects cluster at low pressure and high etch rate.

Discussion & Conclusion

Across 5,000 wafers the defect rate is only 0.14% (7 wafers), yet the defective units share a clear excursion signature: pressure runs about 2.3σ below the mean while temperature and etch rate run roughly 1.4σ high. In other words, the failures concentrate at the low-pressure / high-etch-rate corner of the process window rather than being randomly distributed, which points to pressure control (and its coupling to etch rate) as the first lever for yield improvement. The defect count is too small to support a predictive model, so the conclusion is deliberately limited to the excursion correlation and the control-limit monitoring that would flag such wafers in production.